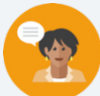


10.4 Solving Problems Involving Arc Length and Area of a Sector

Lesson Introduction

 Benchmark	MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.		 Learning Target	I can solve mathematical and real-world problems involving arc length and sector area.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How can you derive the formula used to find the arc length of a sector using the central angle and circumference of a circle? - How can you derive the formula used to find the area of a sector using the central angle and circumference of a circle?			
 Lesson Narrative	In this lesson, the student will be finding the area of a sector and the length of an arc, respectively, as a fraction of the area and the circumference of the entire circle, using the ratio between the central angle measure and the measure of a whole circle.			
 Component Summary	10.4.1 Warm-Up (~5 min.)	Comparing two deals of pizza prices (<i>SE p. 135</i>)	10.4.4 Collaboration (10 min.)	Discovering the arc length formula (<i>SE p. 137</i>)
	10.4.2 Exploration (10 min.)	Discovering the area of a sector formula (<i>SE p. 136</i>)	10.4.5 Guided Practice (5 min.)	Finding arc length (<i>SE p. 138</i>)
	10.4.3 Guided Practice (5-10 min.)	Solving using area of a sector (<i>SE p. 137</i>)	10.4.6 Guided Practice (5 min.)	Solving using area of a sector and arc length (<i>SE p. 139</i>)
	10.4.7 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p.140</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Area of a Sector - Arc Length <u>Additional Glossary:</u> N/A		(Optional) Large Posters	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may have difficulty isolating the central angle in 10.4.3 Guided Practice.	- Consider having a student write the formulas for finding the area and circumference of a circle on the board for students to use as a reference throughout the lesson. - Consider creating an anchor chart with students as they learn the formulas for Sector Area and Arc Length.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 10.4.2 Exploration positions students to notice the effects of a central angle on the area of the sector in the table provided. Consider applying this routine with different measures of central angles to discover the relationship between the central angle and the area of the sector.	MA.K12.MTR.6.1: Reasonableness Students are required to perform multiple steps to find the area of a sector and the arc length. Due to the problems requiring students to use the order of operations, students may make minor errors that result in an incorrect solution. Encourage students to always check each step of their work, and ask students to verify their solutions by explaining their work.
 Differentiate Instruction	Read and discuss problems aloud to assist students who benefit from auditory entry points. Encourage students to use visuals, such as drawing out circles for students who prefer visual learning methods. Consider allowing students to create anchor charts on larger posters to help kinesthetic and visual learners.	
Lesson Notes:		

10.4.1 Warm-Up: Bell Work

Component Overview: Students will choose between two differently sized pizzas and justify their preferred choice.



10.4.1 Warm-Up: Bell Work

1. A small, family-owned restaurant sells a personal pan pizza with a radius of 10 centimeters for \$5. A nearby restaurant cuts their pizza with a radius of 30 centimeters into 8 slices of equal size, and charges \$5 per slice.

Which is a better deal? Explain your reasoning using mathematics.

Answers may vary. The personal pan pizza has a total area of $100\pi \text{ cm}^2$ for \$5. Meanwhile, the nearby restaurant makes pizzas with an area of $900\pi \text{ cm}^2$. This means that the area of one slice is equal to $\frac{900\pi}{8} = 112.5\pi \text{ cm}^2$. So, for \$5, with an area of $112.5\pi \text{ cm}^2$, the slice is the better deal since you will get more pizza for the same price.



Consider prompting students with questions to help them determine which one is a better deal:

- What is the area of the personal pan pizza?
- What is the area of one slice of the 30 centimeter pizza?
- Would you rather have a pizza with a smaller or larger area?

10.4.2 Exploration: Finding a Formula for the Area of a Sector

Component Overview: Students will work with a partner to discover the formula for the area of a sector by making observations of the fraction of a circle, the central angle, and the equation that can be used to find the area of the sector given.



10.4.2 Exploration: Finding a Formula for the Area of a Sector

- Suppose a circle is cut into 8 equal slices to form sectors.

- Complete the table.

Number of Slices	Fraction of the Circle	Central Angle in Degrees ($^{\circ}$)	Write an Equation to Find the Area of the Sector
2	$\frac{2}{8} = \frac{1}{4}$	$\frac{360^{\circ}}{4} = 90^{\circ}$	$A = \frac{\pi r^2}{4}$
4	$\frac{4}{8} = \frac{1}{2}$	$\frac{360^{\circ}}{2} = 180^{\circ}$	$A = \frac{\pi r^2}{2}$
6	$\frac{6}{8} = \frac{3}{4}$	$\frac{3 \cdot 360^{\circ}}{4} = 270^{\circ}$	$A = \frac{3\pi r^2}{4}$
8	$\frac{8}{8} = 1$	$1 \cdot 360^{\circ} = 360^{\circ}$	$A = \pi r^2$

- Compare the values in your table with your partner. Discuss your equations in the last column and make any revisions if needed.

Students do not need to write an explanation here, but a sample response is provided for reference. For 6 slices, I did not multiply by 3, I only divided by 4.

- Discuss with your partner how the fractions in your equations relate to the central angles, and how the equations might have differed if the angle measure was given in radians.

Students do not need to write an explanation here, but a sample response is provided for reference. The fractions in my equations for the area of the sector are the same as the fraction of the circle's area represented by the central angle, in degrees. If the angle was given in radians, the fraction for the area of the sector would be $\frac{1}{2}$ times the radian measure.



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the table. This will allow students the opportunity to make observations before discussing with a partner.



Encourage students to think of both ways of writing a general formula to find the area of any sector by using degrees and radians.

- What is another unit of measure for angles?
- How could you rewrite your formula to include this unit of measure for angles?
- How would you know when to use which formula?

10.4.2 Exploration: Finding a Formula for the Area of a Sector (continued)

- d. Summarize your discussion by writing a general formula to find the area of any sector.

Answers may vary.

Area of Sector = $\frac{\text{central angle}}{360^\circ} \cdot \pi r^2$, when in degrees.

OR Area of Sector = $\frac{\text{central angle}}{2} \cdot r^2$, when in radians.



Area of a Sector

$$A = \frac{\theta}{360} \pi r^2, \text{ where } \theta \text{ is in degrees}$$

$$A = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

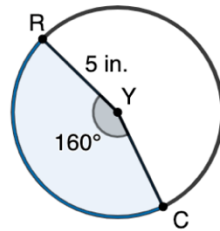
10.4.3 Guided Practice: Area of a Sector

Component Overview: Students will practice finding the area of a sector using the formula.



10.4.3 Guided Practice: Area of a Sector

1. Circle Y has radius $\overline{CY}$ that measures 5 inches with central angle $\angle RYC$, where $m\angle RYC = 160^\circ$. Find the area of the shaded sector. Round your answer to the nearest thousandth.



$$A = \frac{\theta}{360^\circ} \cdot \pi r^2$$

$$A = \frac{160}{360} \cdot \pi (5)^2 = 11.111\pi \text{ in}^2$$

$$\approx 34.907 \text{ in}^2$$

2. Circle X has a diameter of 12 centimeters. Points B and P are on the circle and form the sector BXP . The central angle $\angle BXP$ in Circle X measures 60° . Determine the area of the sector. Express your answer in terms of π .

$$A = \frac{\theta}{360^\circ} \cdot \pi r^2$$

$$A = \frac{60}{360} \cdot \pi (6)^2 = 6\pi \text{ cm}^2$$

3. The area of a sector with a radius of 8 inches is 74.84 square inches. Calculate the approximate measure of the central angle of the sector in degrees.

$$\text{Area of Sector} = \frac{\theta}{360^\circ} \cdot \pi r^2, \text{ with } \theta = \text{central angle.}$$

$$74.84 = \frac{\theta}{360} \cdot \pi (8)^2$$

$$\frac{74.84(360)}{64\pi} = \theta$$

$$\theta \approx 134.0^\circ$$



Consider asking the following:

- How could you find the radius if you were given the area of a sector and the measure of the central angle?
- For question 1, how could you find the area of the non-shaded sector?

10.4.4 Collaboration: Arc Length

Component Overview: Students will work with a partner to make connections with the central angle and the arc length to create a general equation that can be used to find the arc length of a sector.



10.4.4 Collaboration: Arc Length

Work with your partner to complete the following.

1. Explain how the circumference relates to an arc length of a circle.
Arc length is the portion of the circumference. The ratio of arc length to circumference is equal to the ratio of the central angle to the whole circle.
2. Describe how to find the arc length of half a circle.
First, find the circumference of the circle, then divide it by two.
3. Describe how to find the arc length of a quarter of a circle.
First, find the circumference of the circle, then divide it by four.
4. Describe how to find the arc length of a circle with a central angle of 60° .
Find the ratio of the central angle to the whole circle and then multiply that ratio by the circumference of the circle.
5. Write a general equation that can be used to find the arc length of a sector in degrees.

$$\text{Arc Length} = \frac{\text{central angle}}{360^\circ} \cdot 2\pi r$$



For visual learners, consider providing students with a circle with a given circumference. Have students relate the questions to the circle and circumference they were given to help students make connections between the central angle and arc length.

10.4.4 Collaboration: Arc Length (continued)

6. Discuss with your partner how your equation would change if the central angle had been given in radians. Summarize your discussion by writing an equation to find the arc length if the angle measure is given in radians.

I would multiply the radius by the central angle.

I would convert 360° to its radian measure of 2π , then I would divide and cancel out the two 2π 's, leaving the equation:

Arc Length = central angle $\cdot$ r , with the central angle now measured in radians.



Arc Length

Arc Length = $C \cdot \frac{\theta}{360^\circ}$, where θ is in degrees

Arc Length = $r \cdot \theta$, where θ is in radians



For visual learners, consider providing students with a circle with a given circumference. Have students relate the questions to the circle and circumference they were given to help students make connections between the central angle and arc length.

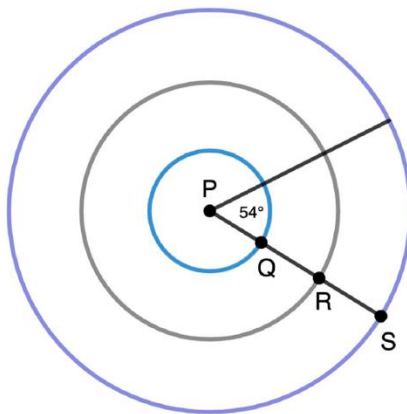
10.4.5 Guided Practice: Arc Length

Component Overview: Students will practice finding the arc length of a sector using the formula.



10.4.5 Guided Practice: Arc Length

- Concentric circles with center P are shown with $m\angle P = 54^\circ$. The smallest Circle P has a radius $\overline{PQ}$ that measures 15 meters (m). The largest Circle P has radius $\overline{PS}$ and the middle Circle P has radius $\overline{PR}$. Each circle has a radius that is 15 m longer than the immediate prior circle.



Calculate the arc length intercepted by $\angle P$ on each circle.

Arc length of circle with radius $\overline{PQ}$	$\begin{aligned} \text{Arc Length} &= \frac{54}{360} \cdot 2\pi(15) \\ &= 4.5\pi \\ &\approx 14.14m \end{aligned}$
Arc length of circle with radius $\overline{PR}$	$\begin{aligned} \text{Arc Length} &= \frac{54}{360} \cdot 2\pi(30) \\ &= 9\pi \\ &\approx 28.27m \end{aligned}$
Arc length of circle with radius $\overline{PS}$	$\begin{aligned} \text{Arc Length} &= \frac{54}{360} \cdot 2\pi(45) \\ &= 13.5\pi \\ &\approx 42.41m \end{aligned}$



Consider asking the following:

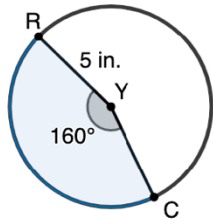
- How are the arc lengths of each sector different if they have the same central angle?
- How could you find the central angle if you were given the radius and the arc length?



Consider challenging students to calculate the arc length in terms of π also.

10.4.5 Guided Practice: Arc Length (continued)

2. Points R and C are on Circle Y . If $m\angle CYR$ is 160° , then find the length of $\widehat{RC}$.



$$\begin{aligned}
 \text{Arc Length} &= \frac{160}{360} \cdot 2\pi(5) \\
 &= \frac{4}{9} \cdot 10\pi \\
 &= \frac{40}{9}\pi \\
 &\approx 13.96 \text{ in}
 \end{aligned}$$



If students found the length of arc RC to be approximately 34.9 in., then they used the area of the circle instead of the circumference.

10.4.6 Guided Practice: Solving Real-World Problems Involving Arc Length and Area of a Sector

Component Overview: Students will calculate arc length and the area of a sector in real-world context.



10.4.6 Guided Practice: Solving Real-World Problems Involving Arc Length and Area of a Sector

1. In Sarah's family, the person celebrating their birthday always gets to cut the first piece of a circular birthday cake. Sarah is celebrating her birthday with both of her parents, her two brothers and her best friend. She cuts her piece of birthday cake and it formed a sector with a diameter of 8 inches and a central angle with measure 85° . If the 5 other party guests equally share the remaining portion of the cake, what is the approximate area of cake each person receives? Round your answer to two decimal places.

First, find the area of the entire cake:

$$\text{Area of Cake} = \pi \cdot 4^2 = 16\pi.$$

Next, find the area of the section of cake cut out by Sarah:

$$\text{Area of Sector} = \frac{85 \cdot 4^2 \cdot \pi}{360} = \frac{34}{9} \pi \text{ in}^2$$

Subtract to get the remaining area:

$$\text{Remaining Area} = 16\pi - \frac{34}{9}\pi = \frac{110}{9}\pi \text{ in}^2$$

Finally, divide the remaining area by 5:

$$\text{Area of each piece of cake} = \frac{110}{9}\pi \div 5 = \frac{22}{9}\pi \text{ in}^2$$

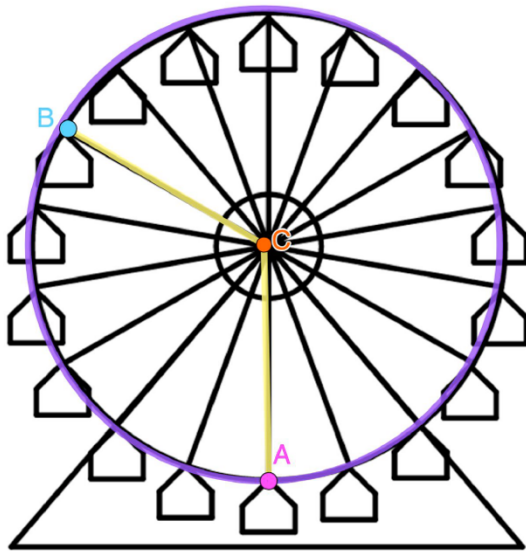
$$\text{Area of each piece of cake} \approx 7.68 \text{ in}^2$$



For students who struggled during the lesson, consider using this time to work with individual or small groups of students. It will be important to determine which part in the process students may be struggling with. Consider having students start the problem to identify where the help is needed.

10.4.6 Guided Practice: Solving Real-World Problems Involving Arc Length and Area of a Sector (continued)

2. Dominic is riding a Ferris wheel at the county fair. The Ferris wheel forms a circle with center C . The Ferris wheel has a radius of 60 ft and rotates clockwise with seats that are equidistant from each other.



Consider reminding students:

- Which way clockwise is.
- That a revolution is one full circle.

- a. After the ride begins from point A it pauses at point B . How many feet has Dominic traveled when his seat pauses at point B ? Write your answer in π form and rounded to the nearest thousandth.

π or feet

- b. About how many feet has Dominic traveled if the full ride is 4 revolutions?

π or feet

10.4.7 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



10.4.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.



Consider providing online resources for additional enrichment opportunities for students to practice or test their abilities.